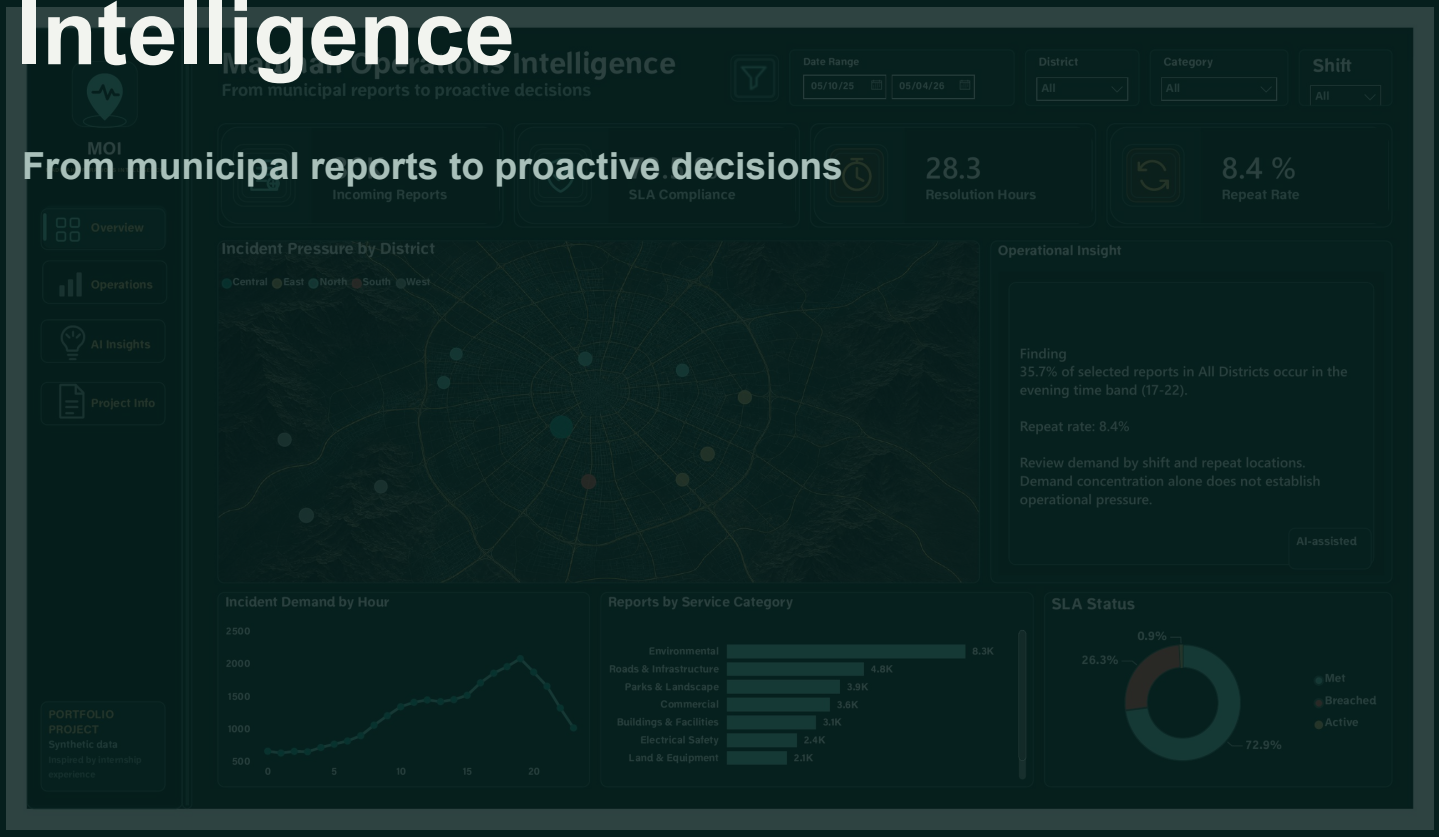
Madinah Operations Intelligence

From municipal reports to proactive decisions



BUSINESS INTELLIGENCE / OPERATIONS ANALYTICS / PREDICTIVE AI

An end-to-end portfolio solution inspired by internship experience at the Municipal Operations Monitoring Center in Madinah.

Yomna Alhejaili

Data Analytics, Business Intelligence & AI-Assisted Operations

PORTFOLIO CASE STUDY / 2026

01 Context & Business Challenge

The opportunity

Municipal operations generate a continuous stream of service reports. The challenge is not only to count them, but to translate demand, capacity and service-level performance into timely operational decisions.

PROJECT FACTS

Synthetic data window
05 Oct 2025 - 05 Apr 2026

Dataset
30,000 synthetic municipal reports

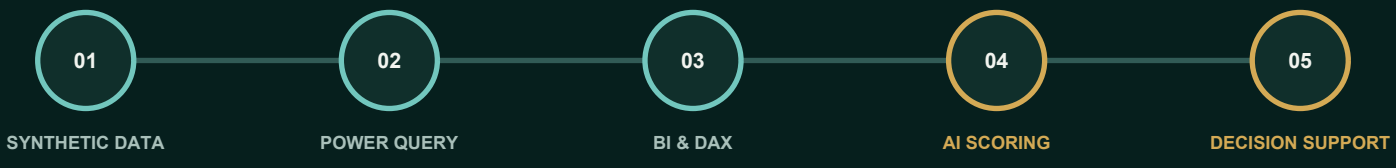
Domain
Municipal Operations Intelligence

Core operational questions

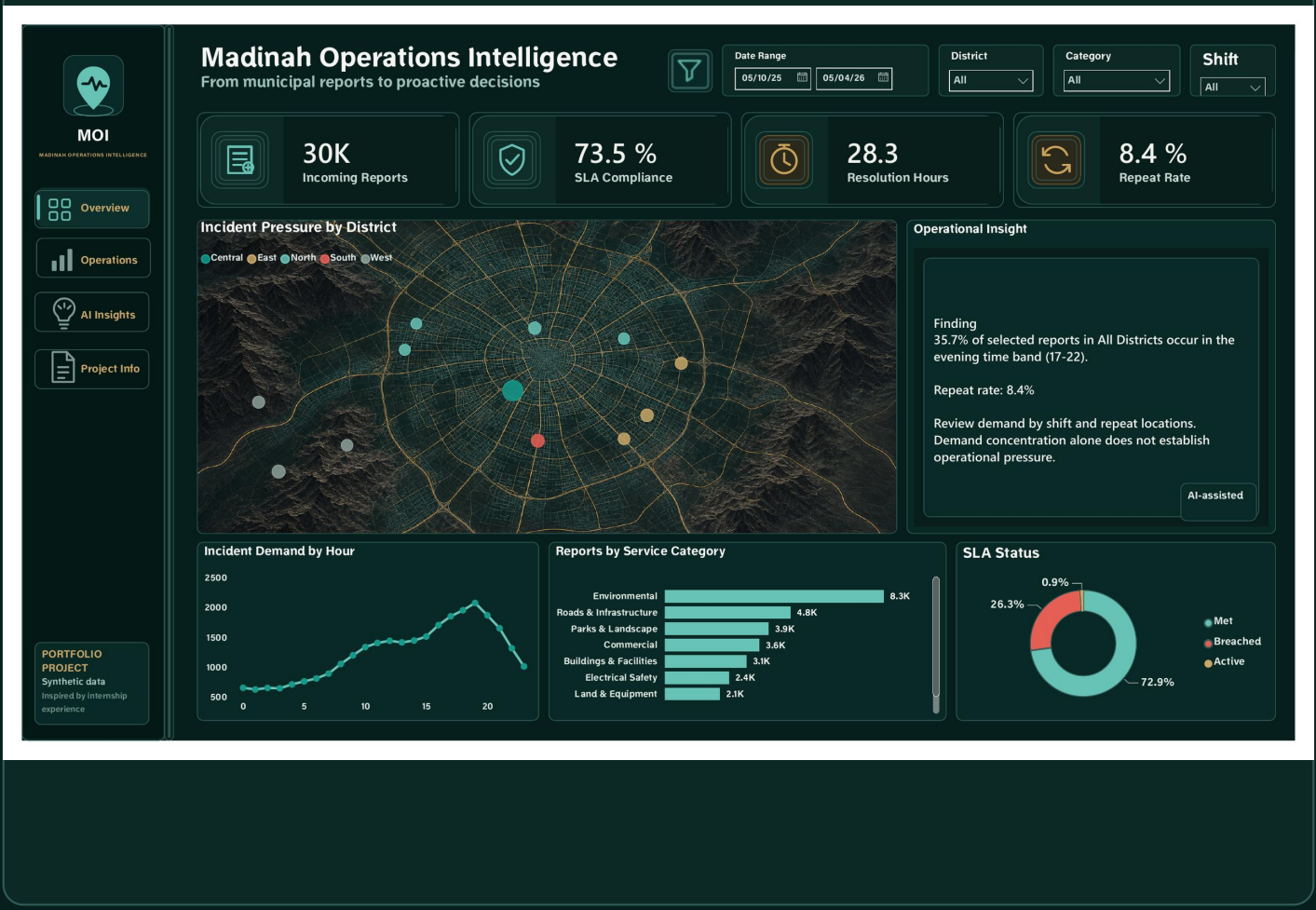
- Where is report demand concentrated?
- How does demand compare with nominal capacity?
- Which districts and shifts are at risk of SLA breach?
- Which active reports should teams review first?

DESIGN PRINCIPLE

Move from descriptive reporting to proactive, human-reviewed decision support.



02 Executive Overview



30K
INCOMING REPORTS

73.5%
SLA COMPLIANCE

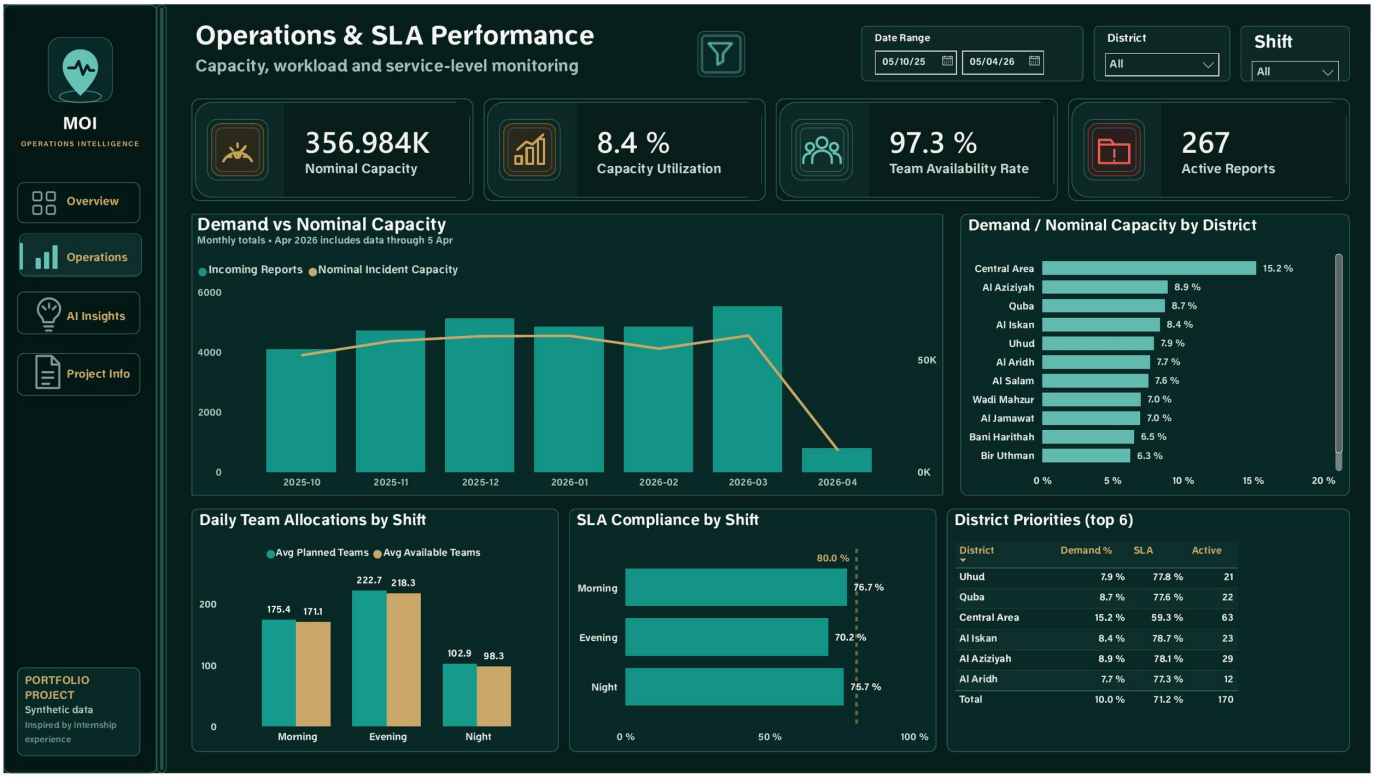
28.3h
MEDIAN RESOLUTION

8.4%
REPEAT RATE

What this page enables

- A single executive view of demand, SLA performance and repeat-report pressure.
- A spatial view that highlights report concentration around the Central Area.
- Hourly and service-category patterns that support resource planning.

03 Operations & SLA Performance



Operational interpretation

- Monthly demand uses nominal synthetic incident capacity (356,984 total).
- District ratios describe synthetic demand; they do not establish overload.
- Shift allocations and SLA outcomes support human review, not staffing prescriptions.

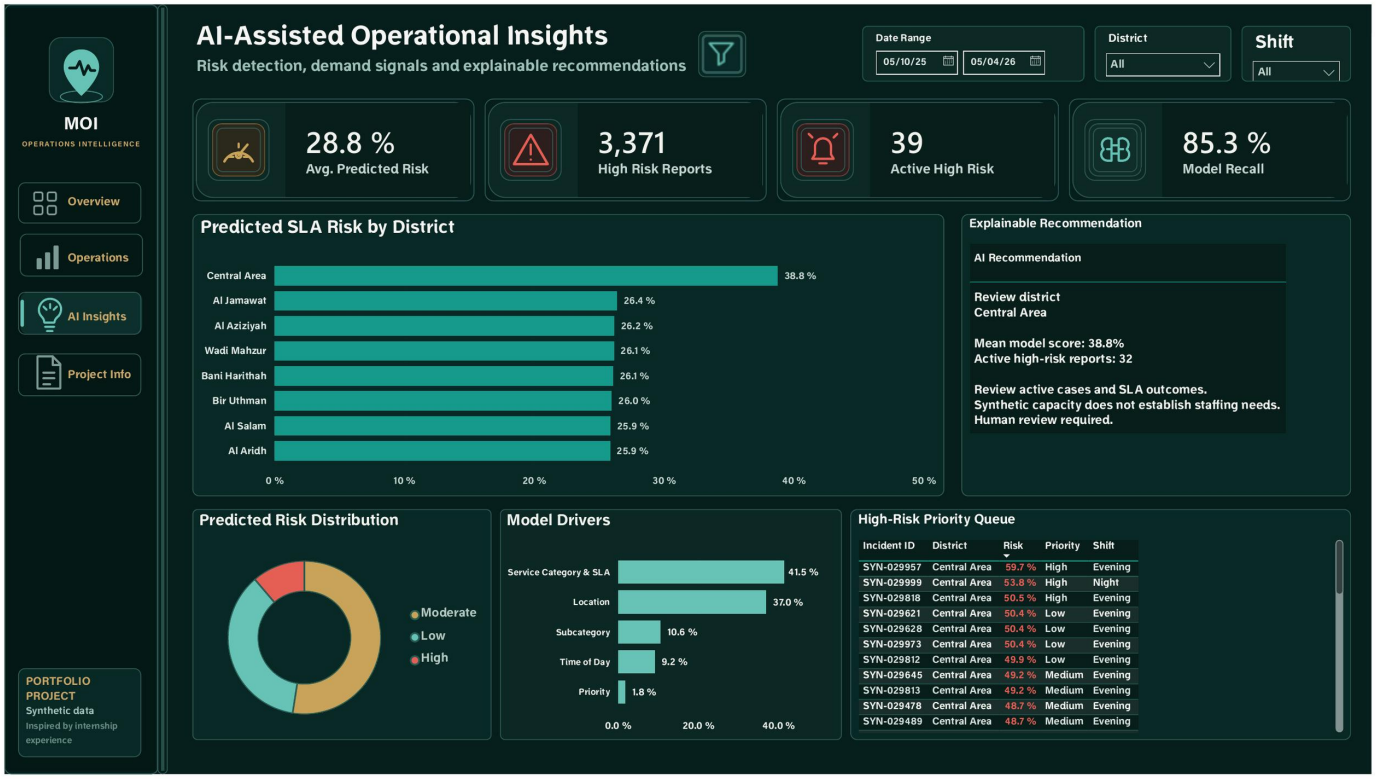
DECISION SIGNAL

Central demand / nominal capacity: 15.23%, while current SLA compliance is 59.3%.

Recommended response

Verify allocations and productivity before making staffing or overload recommendations.

04 AI-Assisted Risk Prioritization



68.7%
TEST ROC-AUC

85.3%
MODEL RECALL

24%
DECISION THRESHOLD

39
ACTIVE HIGH-RISK

MODEL APPROACH

Target: SLA breach before resolution
Models: scikit-learn Logistic Regression + AdaBoost
Pipeline: ColumnTransformer preprocessing
Validation: time cohorts; cutoff-aware labels

EXPLAINABILITY & GOVERNANCE

Precision is 33.6%: 2,861 false alerts on 6,057 resolved test reports. High risk starts at 40%; the review threshold is 24%.
Retrospective evaluation; human review required.

AI-assisted - Human review required

05 Outcome, Governance & Next Steps

Prototype capabilities

01
Visibility
A unified view of demand, location, service category and SLA performance.

02
Operational control
Nominal capacity and shift analysis describe allocations, with explicit limits.

03
Proactive prioritization
Predicted risk helps teams focus on reports most likely to breach SLA.

Responsible use

- All source records are synthetic and privacy-safe.
- The model is a portfolio prototype and is not production-deployed.
- Feature timing and right censoring remain limitations; test results are retrospective.
- Predictions support - and do not replace - human operational decisions.
- A production path would add live integration, monitoring, drift checks and periodic retraining.

ABOUT THE AUTHOR

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